

Chapter 6: Anatomy of Flowering Plants

Dicot Stem

BAIO LEARNING SYSTEM · Narendra Sir — Founder/CEO

Class Notes · Topic 4/8

Hypodermis = collenchyma (mechanical strength)

Ground tissue = cortex + pith, divided by ring of VBs

Vascular bundles in a ring: conjoint, open, endarch

Pericycle = sclerenchyma patches

Medullary rays between vascular bundles

Epidermis

Outermost single layer of barrel-shaped cells, covered with a thin cuticle; may bear multicellular hairs (trichomes) and a few stomata.

Cortex — Hypodermis

Cortex cells are arranged in multiple layers between the epidermis and endodermis, and consist of two sub-zones. **Hypodermis**: a few layers of **collenchymatous** cells just below the epidermis, providing mechanical strength to the young stem.

Cortex — Cortical Layers & Endodermis

Cortical layers (ground tissue): below the hypodermis, rounded thin-walled parenchymatous cells with conspicuous intercellular spaces. **Endodermis**: the innermost layer of the cortex; a single layer of barrel-shaped cells without intercellular spaces, rich in starch grains, so it is also called the **starch sheath**.

Stele — Pericycle

All tissue on the inner side of the endodermis — pericycle, vascular bundles and pith — constitute the stele. The pericycle lies on the inner side of the endodermis, above the phloem, as **semi-lunar patches of sclerenchyma**.

Stele — Vascular Bundles

A large number of vascular bundles are arranged in a **ring** — a characteristic feature of dicot stems. Each vascular bundle is **conjoint, open**, with **endarch** protoxylem (protoxylem towards the centre).

Medullary Rays & Pith

Between the vascular bundles lie a few layers of radially placed parenchymatous cells, called **medullary rays**. The **pith** is a large number of rounded parenchymatous cells with large intercellular spaces

occupying the central portion of the stem.

T.S. of Dicot Stem – Regions (outside → inside)

Region	Key Feature
Epidermis	Cuticle, trichomes, a few stomata
Hypodermis	Collenchyma — mechanical support
Cortical layer	Parenchyma, intercellular spaces
Endodermis	Starch sheath
Pericycle	Sclerenchyma patches
Vascular bundle	Ring arrangement, conjoint, open, endarch
Pith	Large, central, parenchymatous

4 Main Regions of Dicot Stem T.S.

- Epidermis
- Cortex (hypodermis + cortical layer)
- Endodermis (starch sheath)
- Stele (pericycle + ring of vascular bundles + pith)